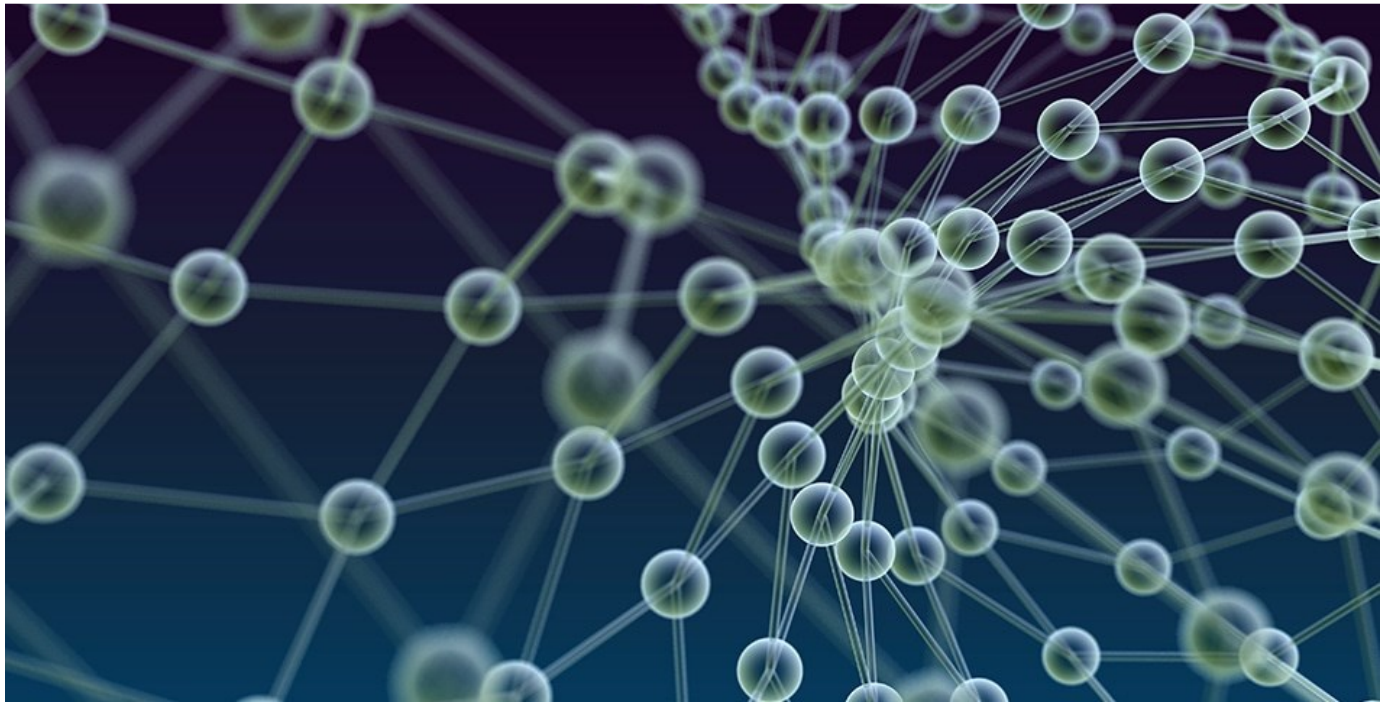
Smart Polymers:-

Smart polymers, also known as stimuli-responsive or intelligent polymers, are materials that can undergo reversible changes in their physical or chemical properties in response to external stimuli such as temperature, pH, light, electric field, or solvent composition.

These materials have attracted significant attention due to their potential applications in various fields including drug delivery, tissue engineering, sensors, actuators, and controlled release systems.



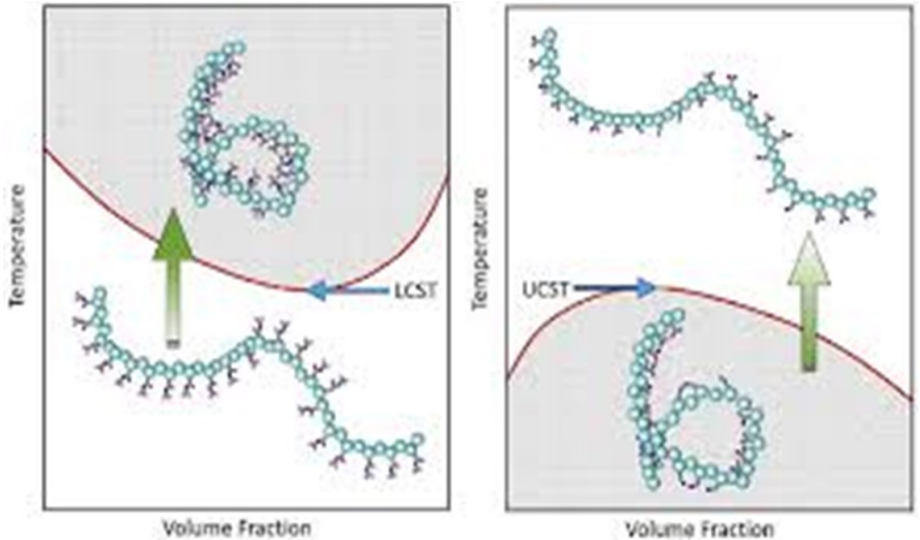
Temperature responsive polymers:-

- Thermosensitive polymers are a class of intelligent materials capable of adjusting their properties in response to temperature variations.

- They achieve this by leveraging the controlled and easily measurable stimulus of temperature, which triggers the expansion or contraction of their polymer chains.

- The crucial factor in this transition lies in the equilibrium between the hydrophilic (water-attracting) and hydrophobic (water-repelling) segments along the polymer chain, coupled with the overall energy dynamics of the system.

- In the case of lower critical solution temperature (LCST polymers), they dissolve readily at lower temperatures but become insoluble as the temperature rises. Conversely, UCST polymers dissolve at higher temperatures yet lose solubility at lower temperatures.

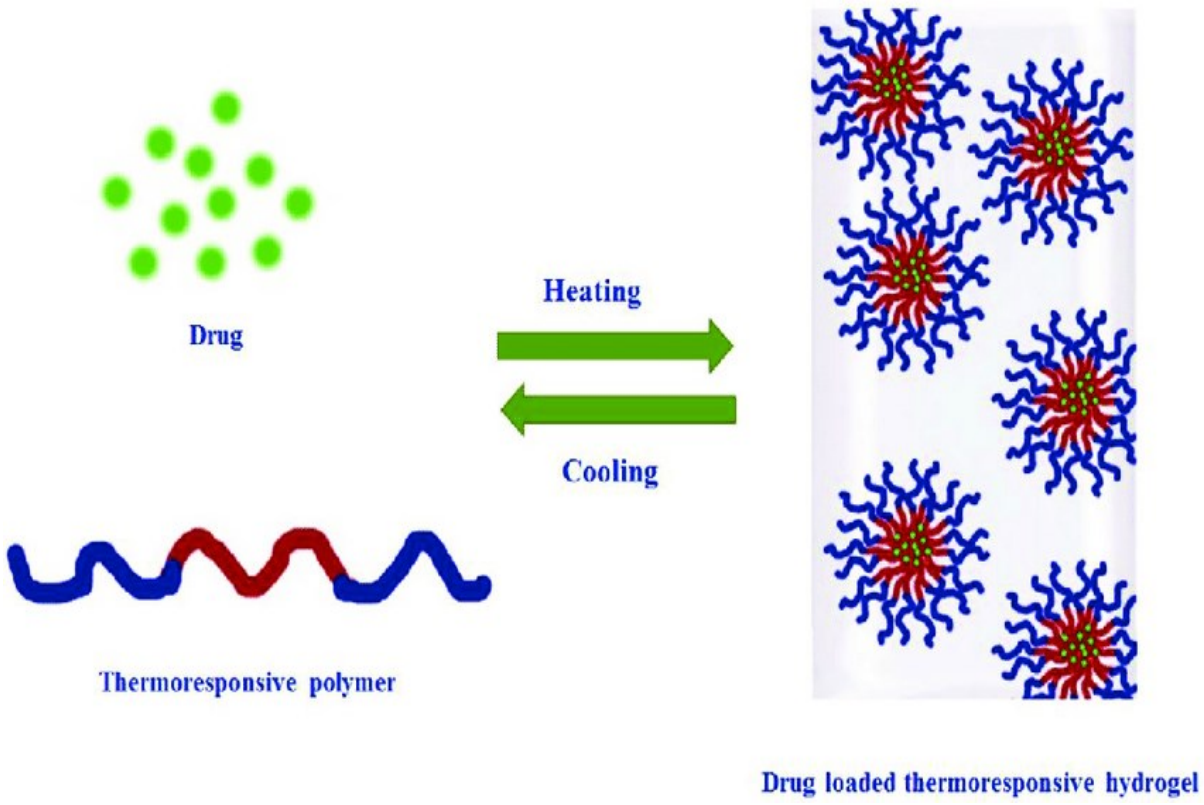


Examples

LCST: Poly (vinyl amide), poly (N-substituted acrylamide), poly (N vinylcaprolactam), cellulose, chitosan, xyloglucan and PLGA–PEG– PLGA triblock copolymers

UCST: poly (2-dimethyl) methacryloxyethyl-ammonium propanesulfonate

Applications:



Drug Delivery

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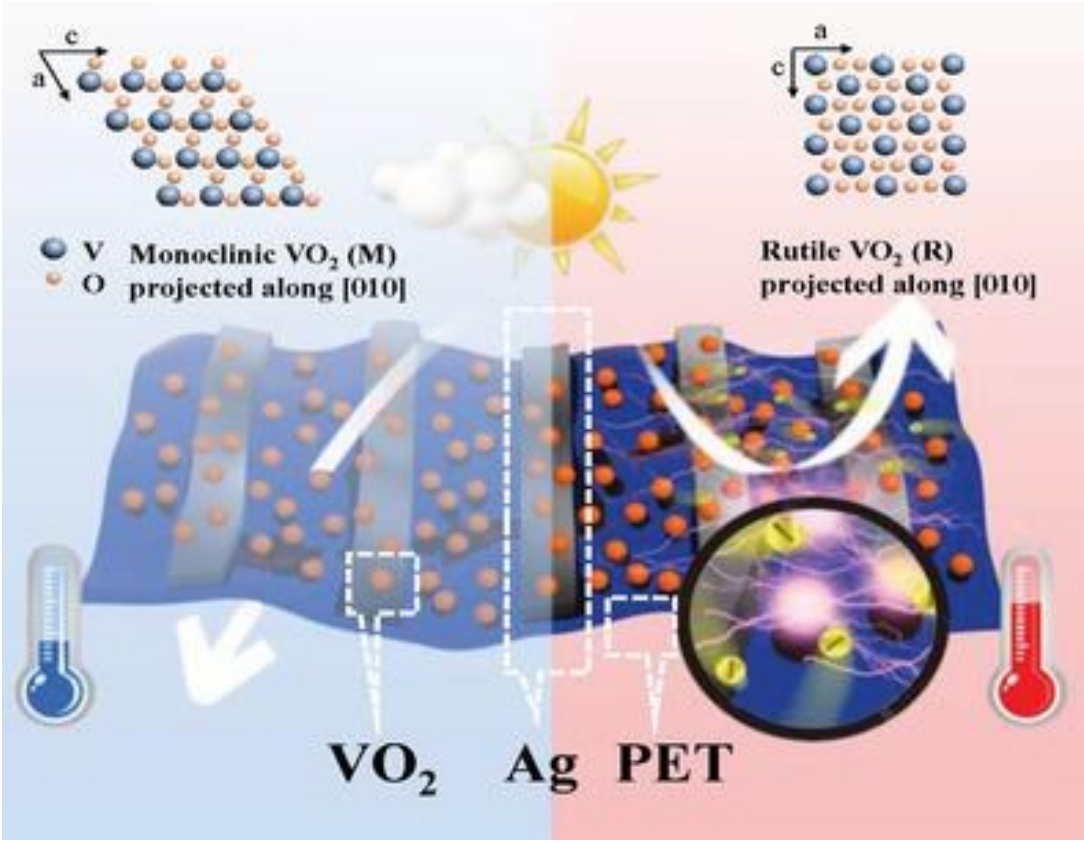

Smart Textiles

https://xpic.x-mol.com/20200114/10.1002_admt.201900599.jpg

Electro-responsive polymers:-

- Electro-responsive or electroactive polymers (EAPs) represent a specialized category within smart polymers capable of dynamically adjusting their physicochemical characteristics upon exposure to electric signals.

- Through the conversion of electrical energy into mechanical energy, they exhibit reversible changes in shape, including swelling, shrinking, or bending. These alterations depend on various parameters of the electric current, such as its magnitude, duration, and frequency.

Example:- polythiophene, polypyrrole,

Mechanism of working of Electro-responsive polymers:

- **Conductivity Change:** Some electroresponsive polymers change their electrical conductivity when subjected to an electric field. This can happen through doping (adding or removing electrons), which alters the polymer's electronic structure, making it more or less conductive.
- **Electrostriction:** This refers to the change in the shape or size of the polymer when an electric field is applied. The polymer may expand, contract, or bend in response to the electric field, making it useful in applications like actuators or artificial muscles.
- **Ion Migration:** In some polymers, the application of an electric field causes ions within the material to move, leading to changes in the material's properties. This movement can cause swelling, deformation, or changes in conductivity.

Applications.

Actuators and Artificial Muscles

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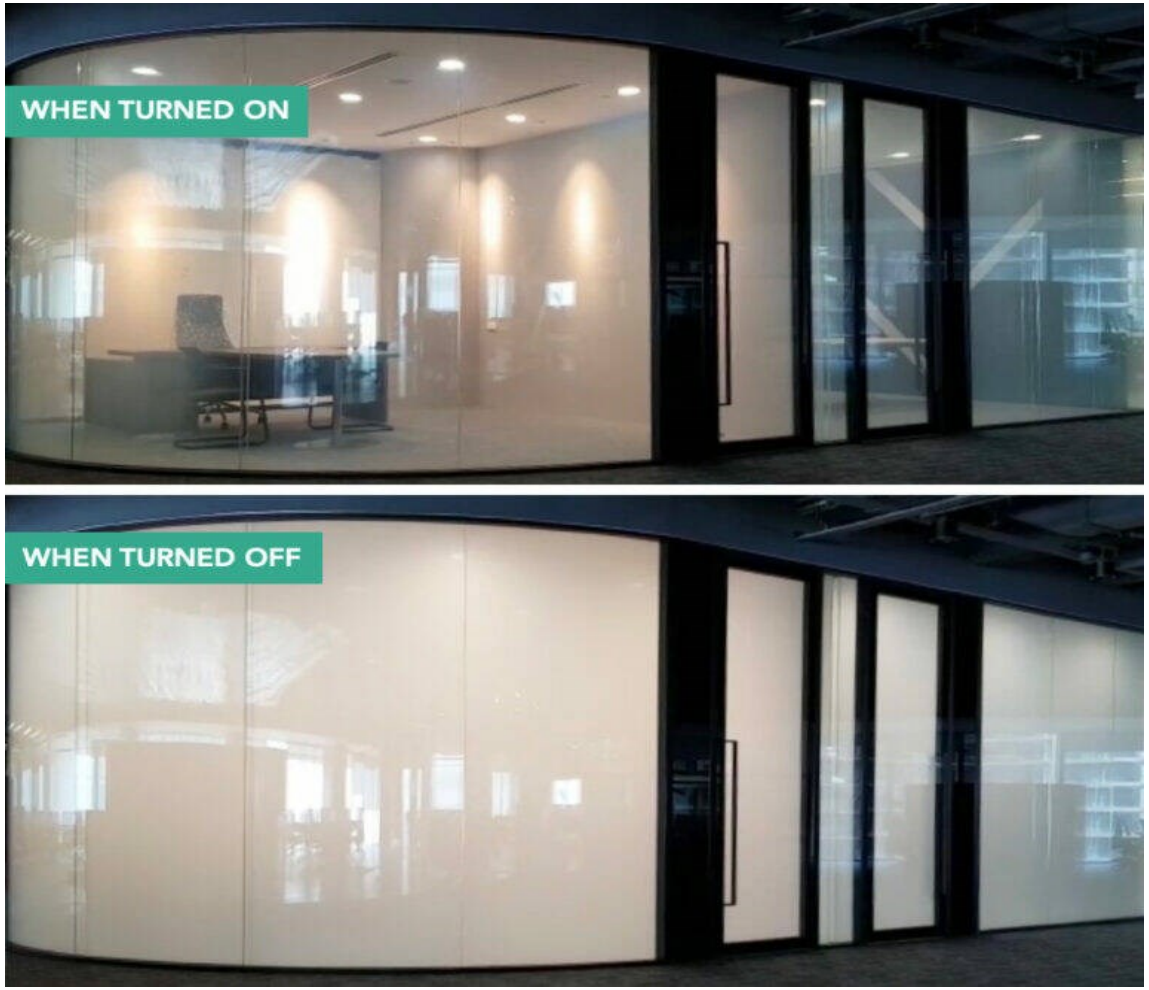


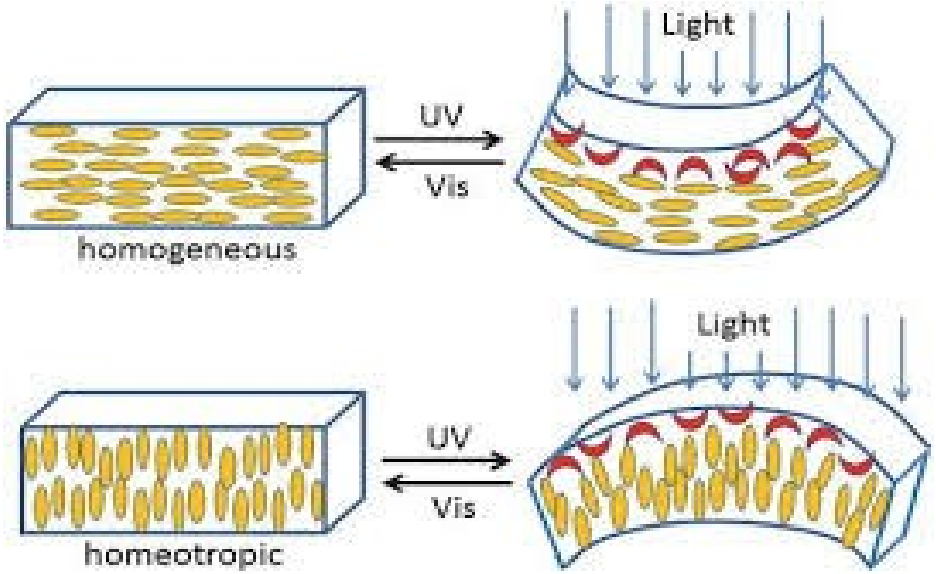
Smart Windows and Coatings

https://www.radiantvisionsystems.com/sites/default/files/styles/d12/public/2022-03/switchable%20glass_office.jpg.webp?itok=jCPxsWXX

Photo-responsive polymers

Photo-responsive polymers are materials that undergo changes in their physical or chemical properties when exposed to light.

These changes can include alterations in shape, color, solubility, or mechanical properties, and they are typically reversible, meaning the polymer can return to its original state when the light stimulus is removed or altered.



Mechanism of working of Photo-responsive polymers:

Photoisomerization:

- **Mechanism:** This is one of the most common mechanisms in photoresponsive polymers. Certain molecules within the polymer structure, known as chromophores, can switch between two different forms (isomers) when exposed to specific wavelengths of light. The most well-known example is azobenzene, which can switch between a cis and trans configuration upon exposure to UV or visible light.
- **Effect:** This change in molecular configuration can lead to macroscopic changes in the polymer, such as bending, stretching, or changes in optical properties. The process is reversible, so the polymer can switch back to its original state when exposed to a different wavelength of light.

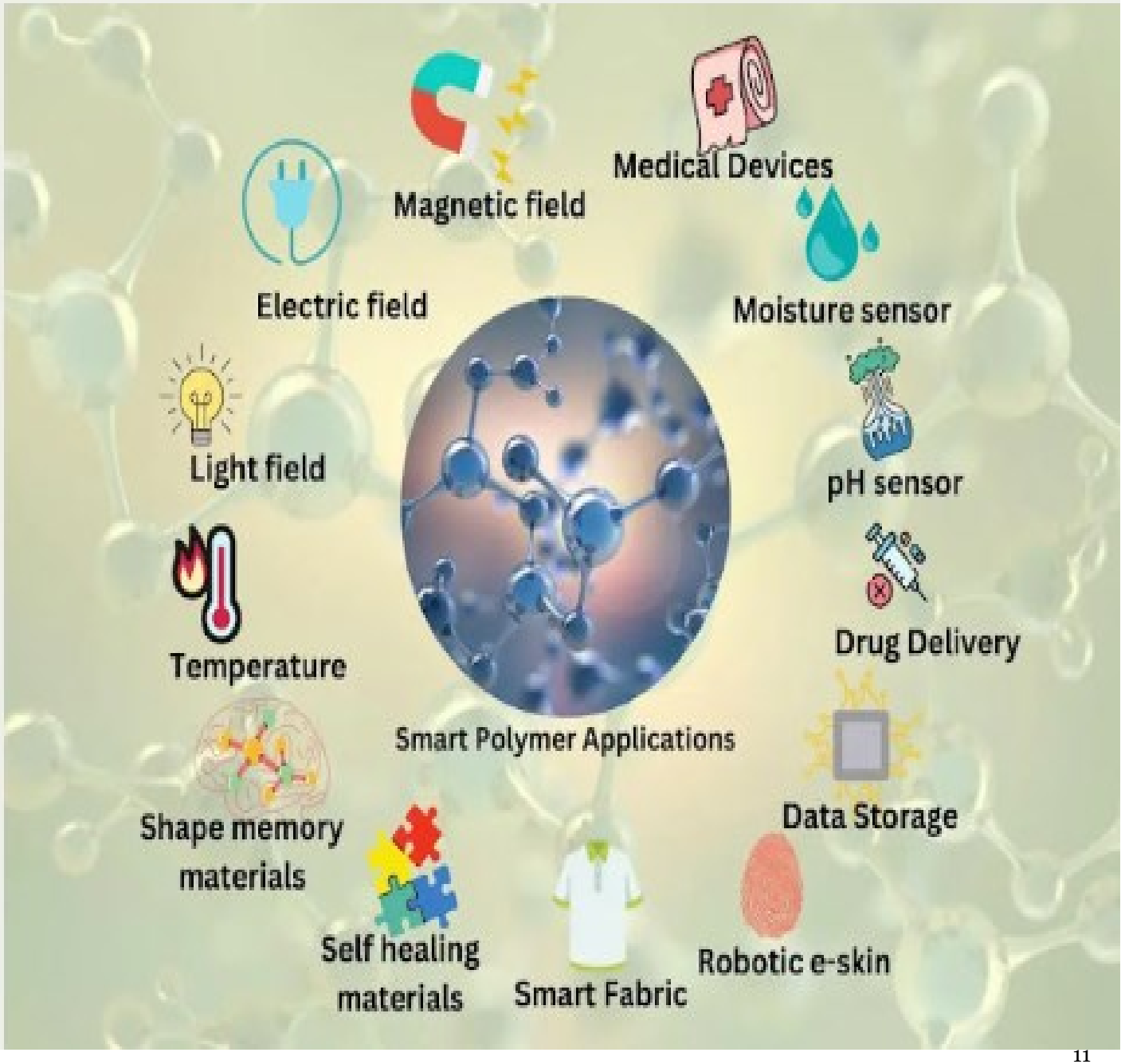
Applications

Smart Coatings

Soft Robotics and Actuators

Smart polymers -Applications:-

- Smart polymeric carriers can deliver drugs in response to a stimulus.
- Smart polymers can be engineered to detect and respond to specific biomolecules, such as disease markers or drugs.
- In addition to this, various types of smart polymers can be utilized across different fields



Filtration Membranes

Overview of the TOPIC

- Filtration Membranes
- Types of Filtration Membranes
- Applications of Filtration Membranes

Importance of Filtration Membranes

- **Health Benefits:** Filtration membranes significantly reduce pathogens, ensuring safer drinking water and food.
- **Industrial Efficiency:** Integration of membranes enhances production efficiency by optimizing resource usage in various processes.
- **Waste Reduction:** Utilizing membranes minimizes waste pollutants, promoting sustainable practices across industries.

Advantages of polymers in Filter membranes

- Flexibility: More flexible than inorganic membranes.
- Ease of Handling: Simpler manufacturing and processing.
- Cost: Less expensive than inorganic materials like oxides, metals and ceramics.

Sustainable Polymers for Filtration Membranes

- **Definition of Sustainable Polymers:** Sustainable polymers are biodegradable materials that minimize environmental harm.

  Examples: PLA and PHA

- Generally, Water Filtration membranes are made up of polymers which is sustainable in nature.

Introduction to Filtration Membranes

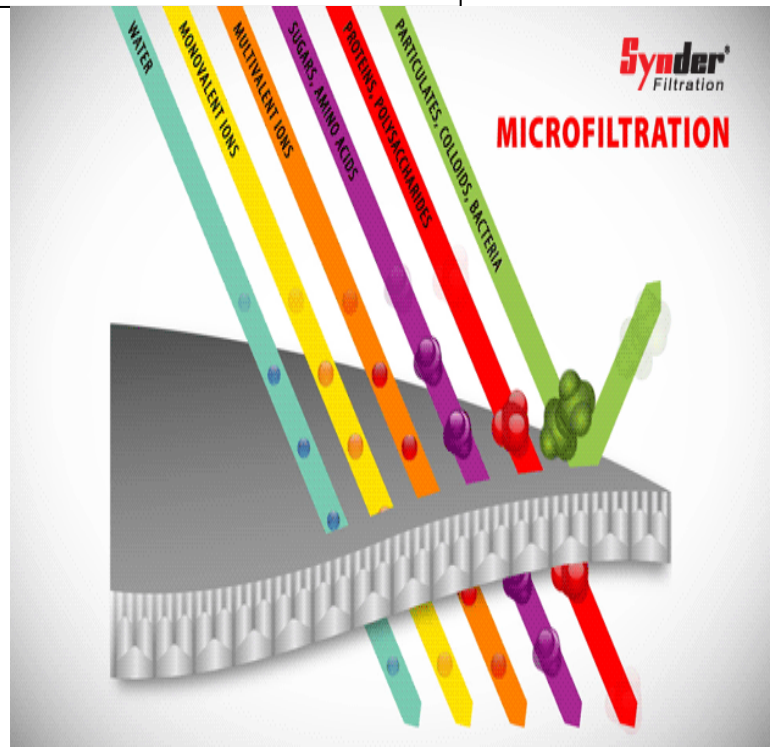
- Membrane filtration is a physical procedure for separation of particles using semi-permeable membranes.
- Membrane filtration is a rapidly expanding field in water treatment.
- Many different types of filters are available in a wide range of pore sizes and configurations.

Types of Filtration Membranes

- Microfiltration
- Ultrafiltration
- Nanofiltration
- Reverse Osmosis

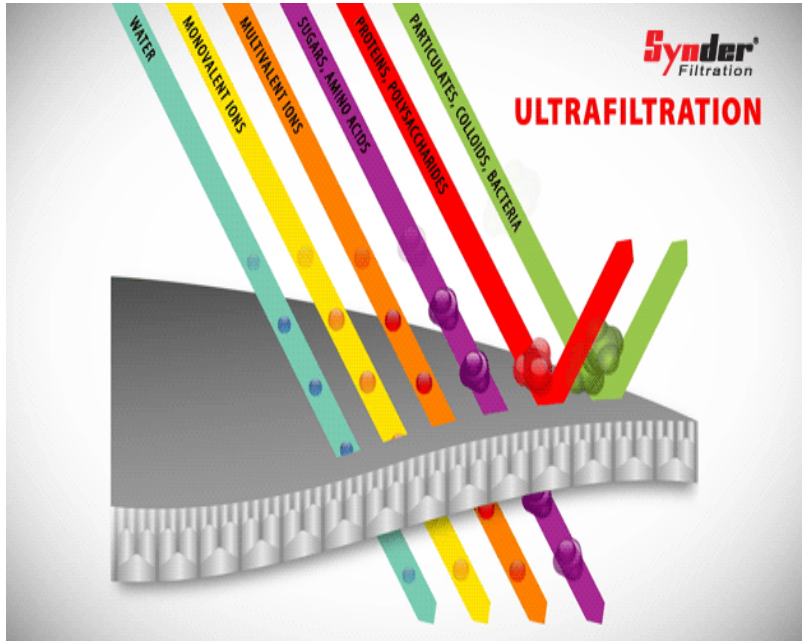
Microfiltration Membranes

- **Pore Size:** Microfiltration membranes typically possess pore sizes ranging from 0.1 to 10 micrometers.
- **Common Materials:** Polymers like PVDF(Polyvinylidene Fluoride) and PES (Polyethersulfone) are favored for their chemical resistance and durability.
- **Removal Capabilities:** Effective in separating large suspended solids such as colloids, particulates, fat, and bacteria while allowing sugars, proteins, salts, and low molecular weight molecules to pass through the membrane.

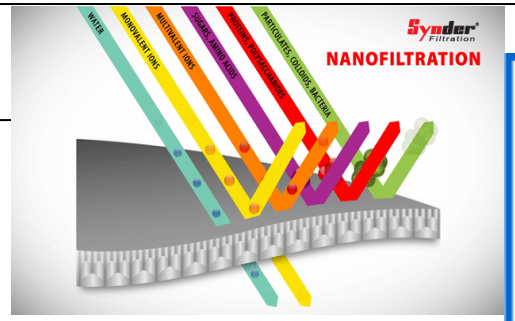


Ultrafiltration membranes

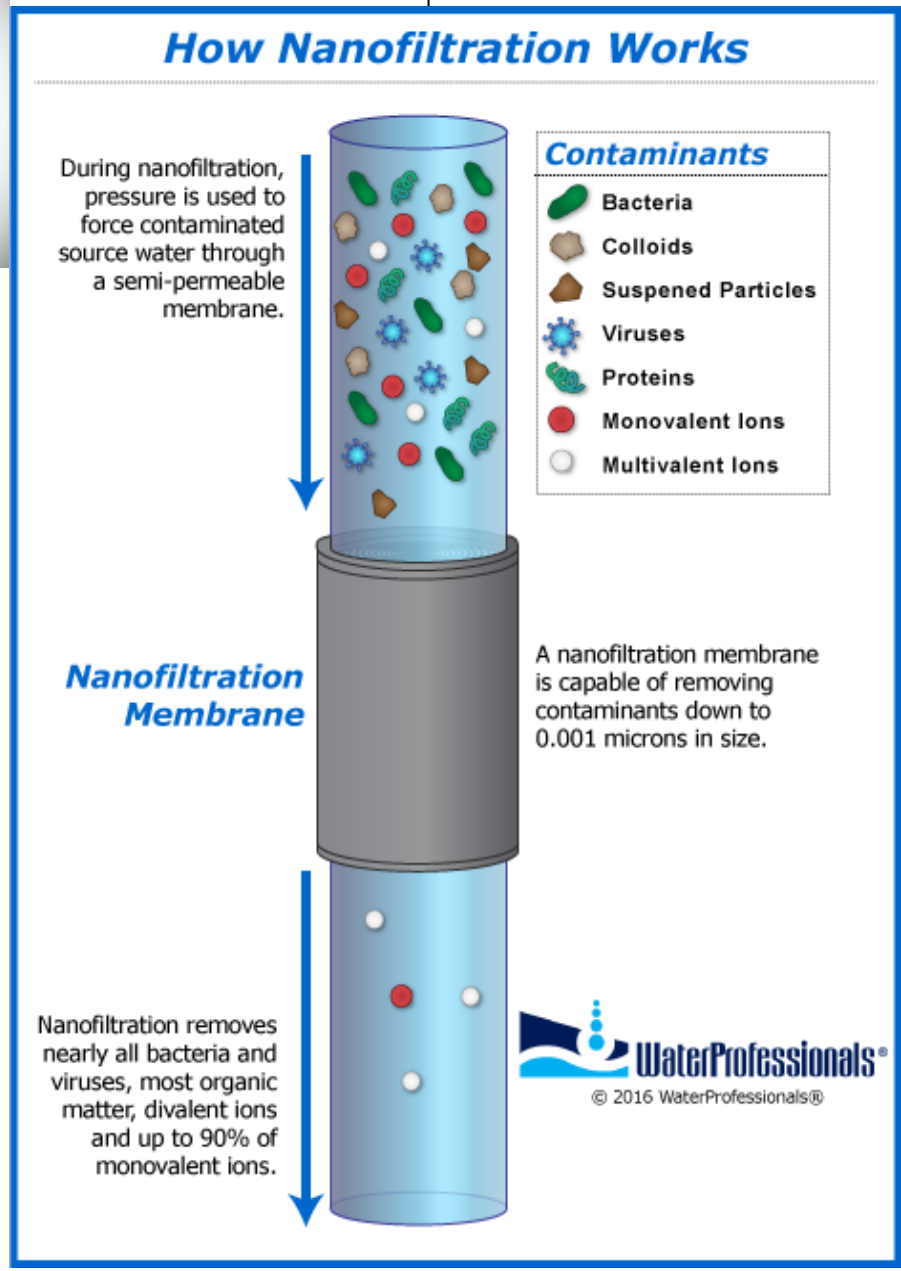
- These membranes have pore sizes of 0.01 to 0.1μm, between that of nanofiltration and microfiltration.
- Ultrafiltration membranes are capable of separating larger materials such as colloids, particulates, fats, bacteria, and proteins while allowing sugars and other low molecular weight molecules to pass through the membrane
- Eg: PES(Polyethersulfone)



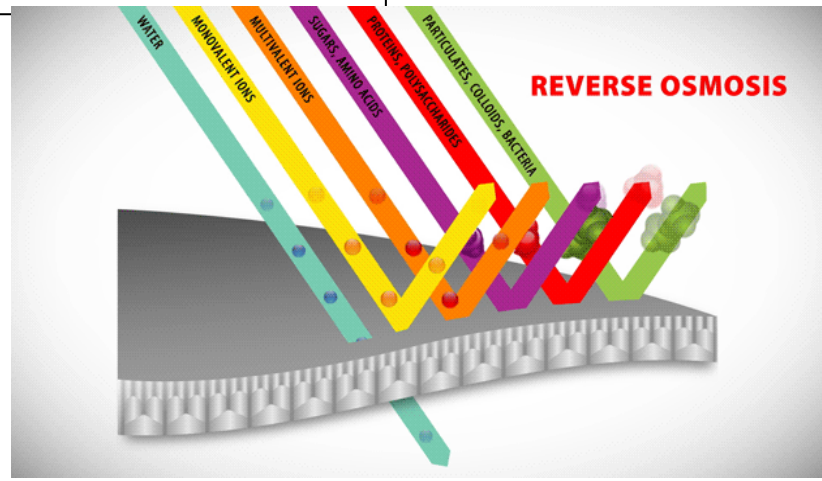
Nanofiltration Membranes



- Nanofiltration membranes feature smaller pore sizes range from 0.1 to 10nm.
- Nanofiltration membranes allow water and some salts to pass through the membrane while retaining multivalent ions, low molecular weight molecules, sugars, proteins, and other organic compounds.
- Nanofiltration effectively eliminates organic compounds, improving water purity and safety.



Reverse Osmosis Membranes

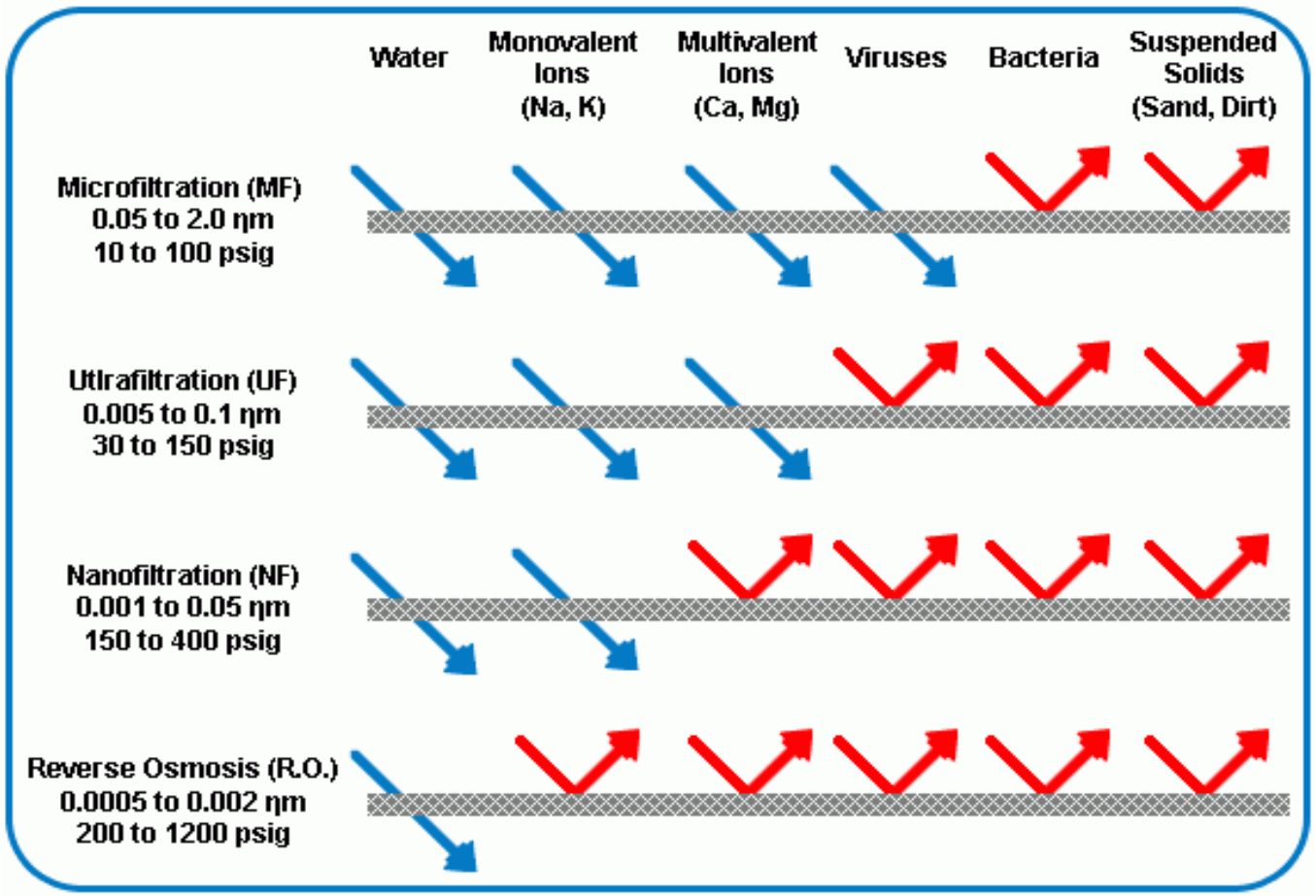


- **Pore Sizes in RO Membranes:** RO membranes typically feature pores smaller than 0.001 micrometers, enabling ion rejection.
- Reverse osmosis is one of the oldest and **most popular separation techniques used mainly for the purification of water.**
- Reverse osmosis is a special type of filtration that uses a semi-permeable, porous membrane, that allows only pure water to pass through it, filtering the larger molecules or impurities.

- RO membranes are in the form of flat sheets and hollow fine fibers.
- The semi-permeable membranes used for the RO process are typically made of a thin polyamide layer (<200 nm) deposited on top of a polysulfone porous layer (about 50 microns) on top of a non-woven fabric support sheet.

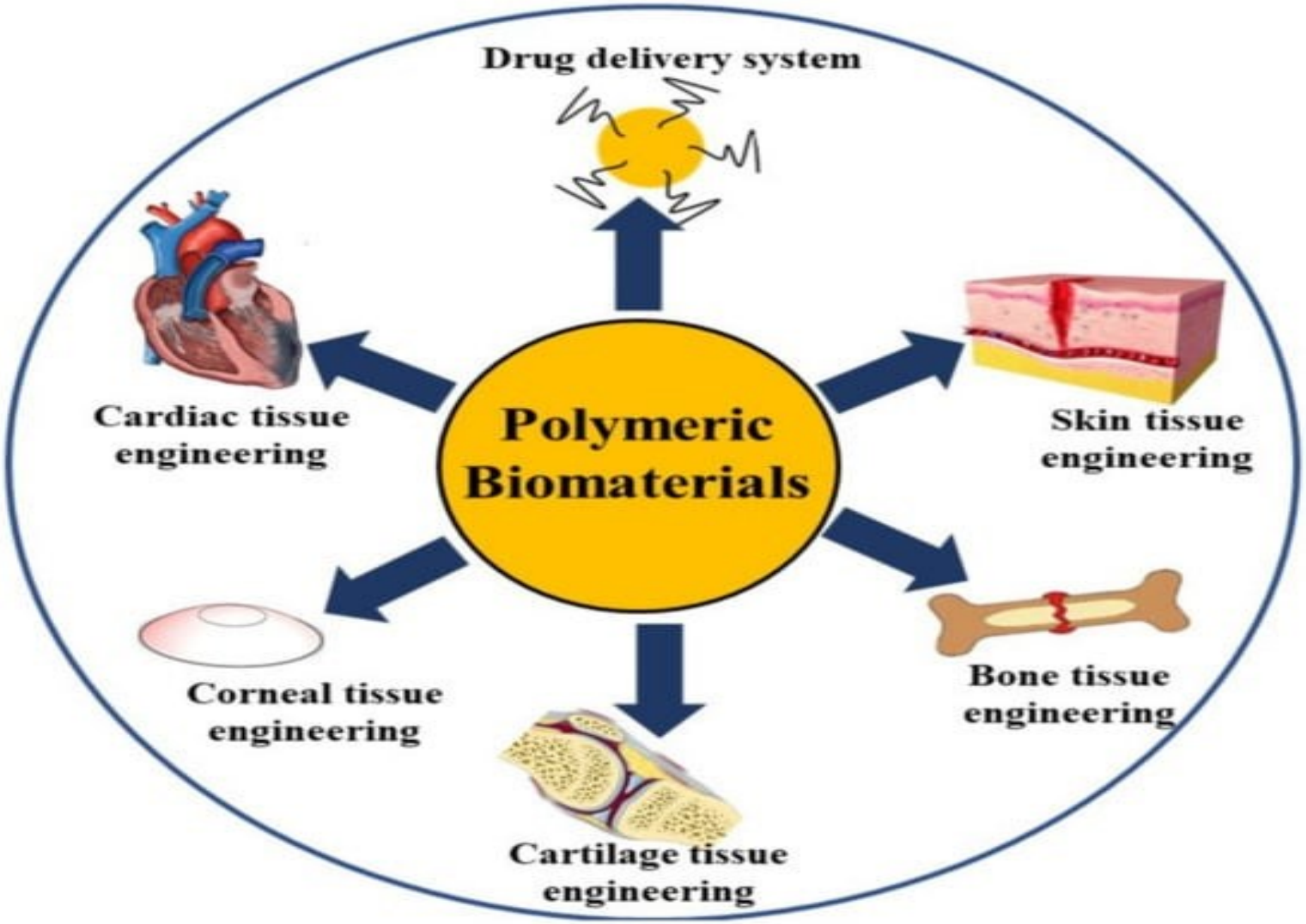
Differences between Filtration membranes

Particular	Microfiltration	Ultrafiltration	Nanofiltration	Reverse Osmosis
Pore size	0.1 to 10 micrometers	of 0.01 to 0.1µm	0.1 to 10nm	<0.001 micrometers
Typical solution treatment	Solution with solid particles	Solution with colloids/macro molecules	small molecules, polyvalent ions	Ions, low molecular weight species, salts
Applied pressure for the process	0.2-5 bar	1-10 bar	5-10 bar	10-150 bar



Schematic Representation for all types of filtration membranes

Type of Filter membranes	Advantages	Disadvantages	Application
Microfiltration	Simple operation and good stability	Cannot remove dissolved salts and small organic molecules	Drinking water pretreatment, primary filtration during water treatment
Ultrafiltration	High efficient removal of macromolecules and colloids	Limited separation on dissolved salts and small organic molecules	Industrial waste water treatment
Nanofiltration	Low energy consumption,High water production rate,significant effect on improving quality	Poor selectivity for monovalent ions and membrane is easily get contaminated	Drinking water softening, organic matter removal
Reverse Osmosis	High water quality,suitable for high purification requirements	High energy consumption ,large amount of waste water	Sea water desalination, pure water preparation, industrial waste water treatment


Drug delivery system
Cardiac tissue engineering
Polymeric Biomaterials
Skin tissue engineering
Corneal tissue engineering
Bone tissue engineering
Cartilage tissue engineering

Polymeric Biomaterials

- Polymeric biomaterials are materials used in medical applications that are composed of polymers, which are long chains of repeating molecular units.
- Polymeric biomaterials play a critical role in various medical devices and applications due to their versatility, biocompatibility, and the ability to tailor their physical and chemical properties for specific uses.

Unique properties of polymers

- Flexibility
- Resistance to biochemical attack
- Good biocompatibility
- Lightweight
- Available in a wide variety of compositions with adequate physical and mechanical properties
- Can be easily manufactured into products with the desired shape.

Thus, due to these features, polymers will find a broad range of applications in the medical field.

Types of Polymeric Biomaterials

1. **Synthetic polymeric biomaterial:** Synthetic polymeric biomaterials are man-made materials created from polymers specifically engineered for use in medical and biological applications
2. **Natural Polymeric biomaterial:** Natural polymeric biomaterials are materials derived from natural sources that are used in medical and biological applications.

Generally, polymeric biomaterials are used extensively in various medical devices, drug delivery systems, and tissue engineering applications due to their versatility, durability, and tunable properties.

Synthetic polymeric biomaterial-Examples

Polyvinylchloride (PVC):

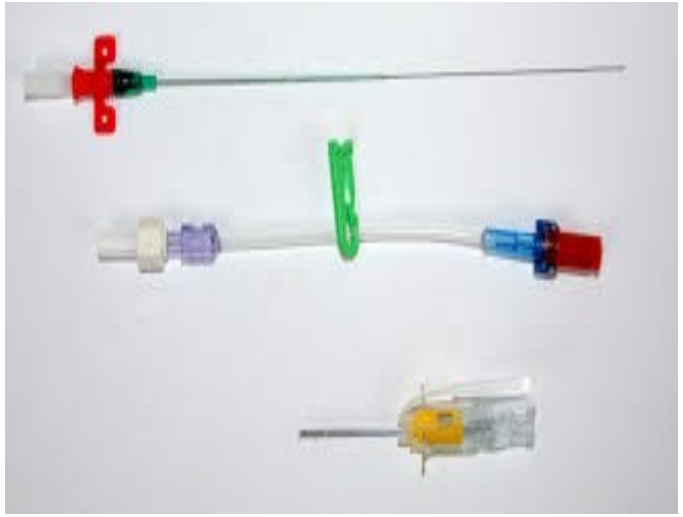
PVC is an amorphous, rigid polymer due to the presence of chloride with a Tg of 75~105°C.

Uses

It is used in blood, solution storage bags, and surgical packaging. PVC is commonly used in dialysis devices, catheters, and cannulae.

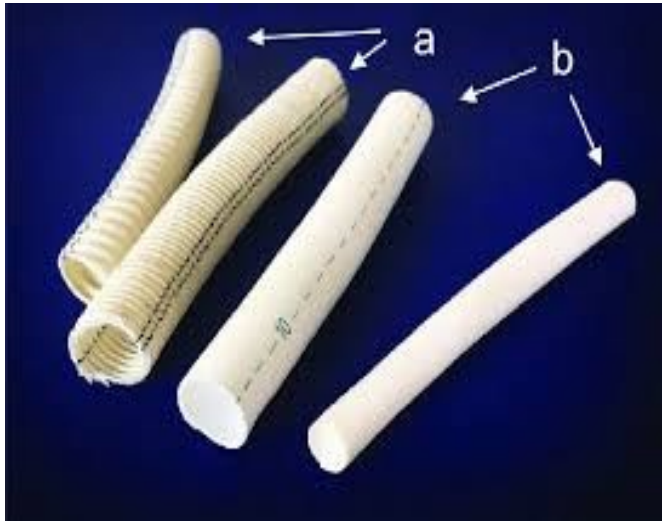
Polyethylene :

Polyethylene (PE) is used in Pharmaceutical bottles, nonwoven fabric, catheters, pouches, flexible containers, and orthopedic implants



Polypropylene

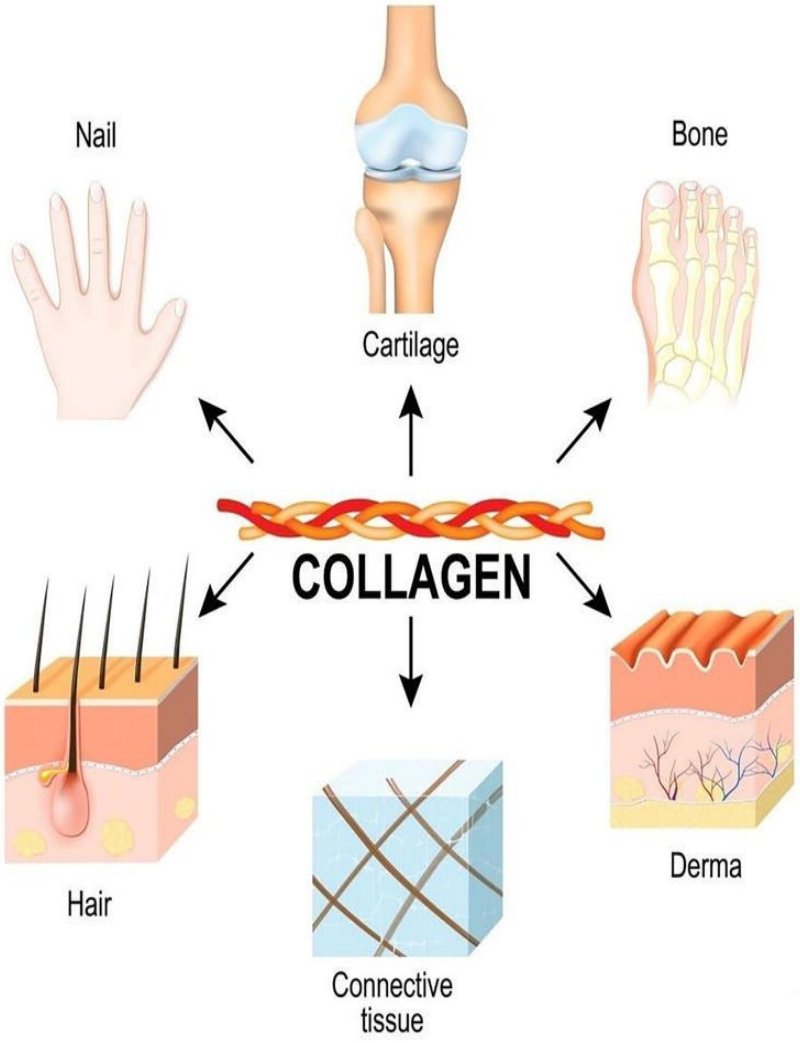
It is used in Disposable syringes, blood oxygenator membranes, sutures, nonwoven fabric, and artificial vascular grafts



Natural Polymeric biomaterial

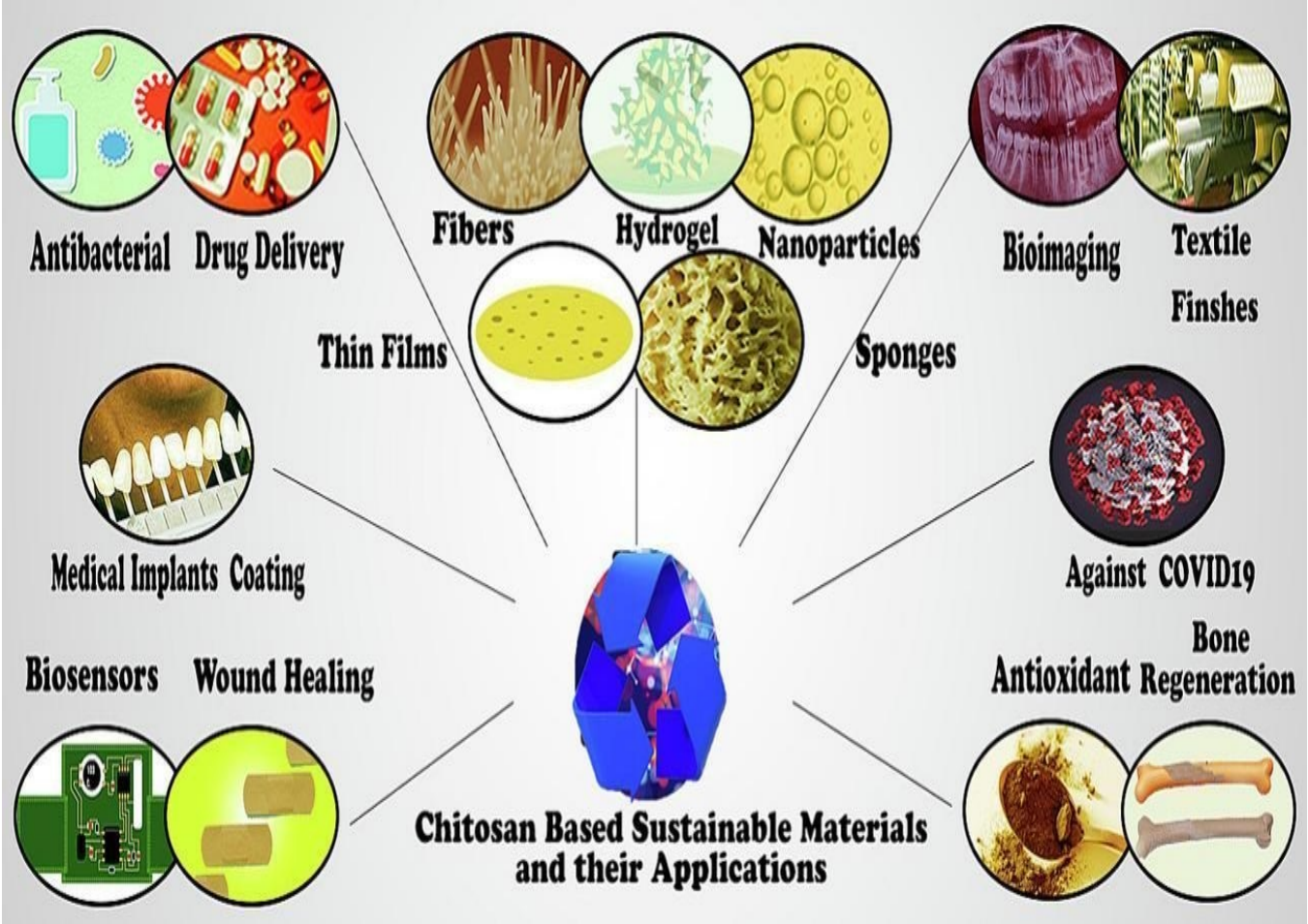
Collagen:

It is a major component of our skin and bones and is a popular choice for wound dressings, drug delivery, and tissue engineering.



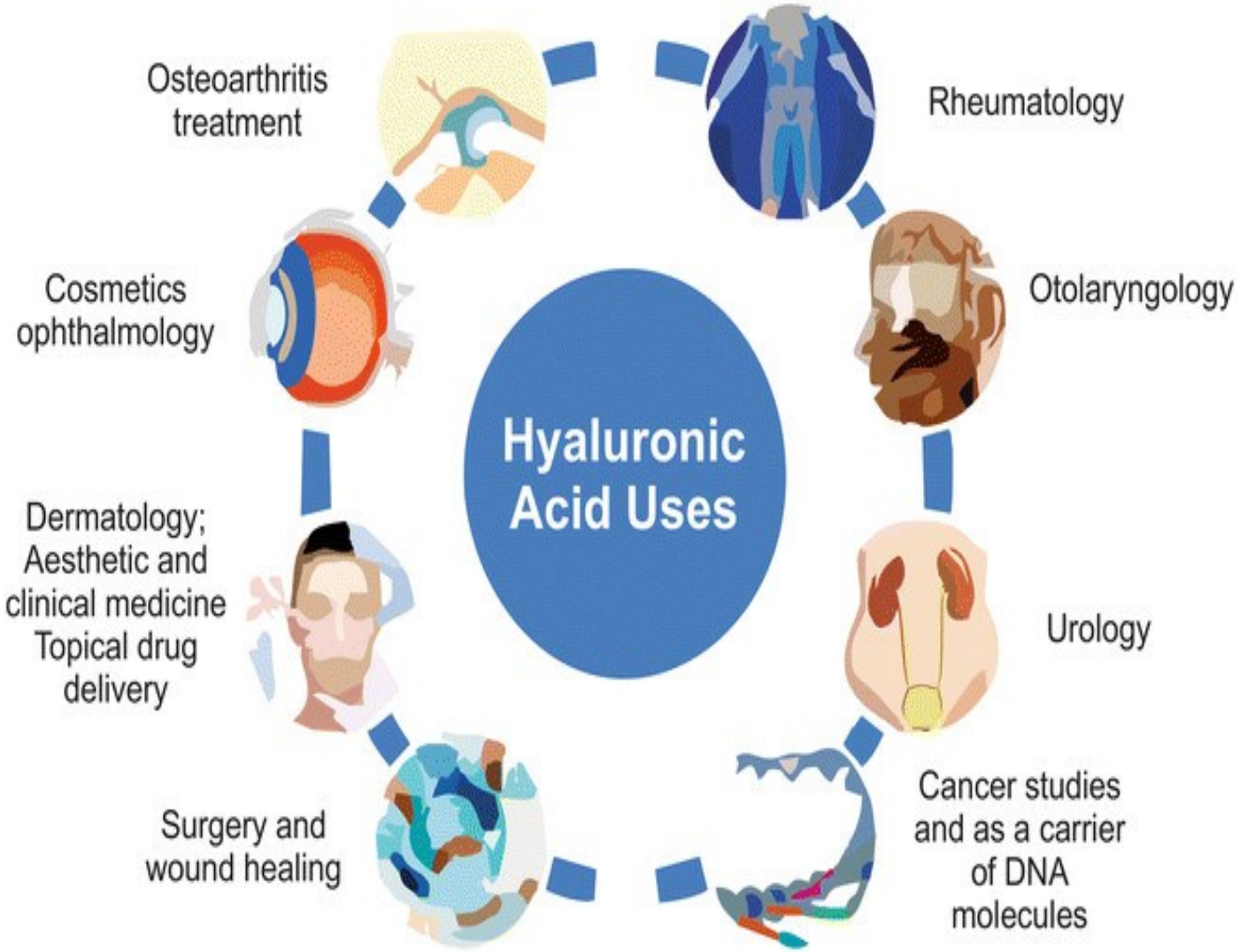
Chitosan:

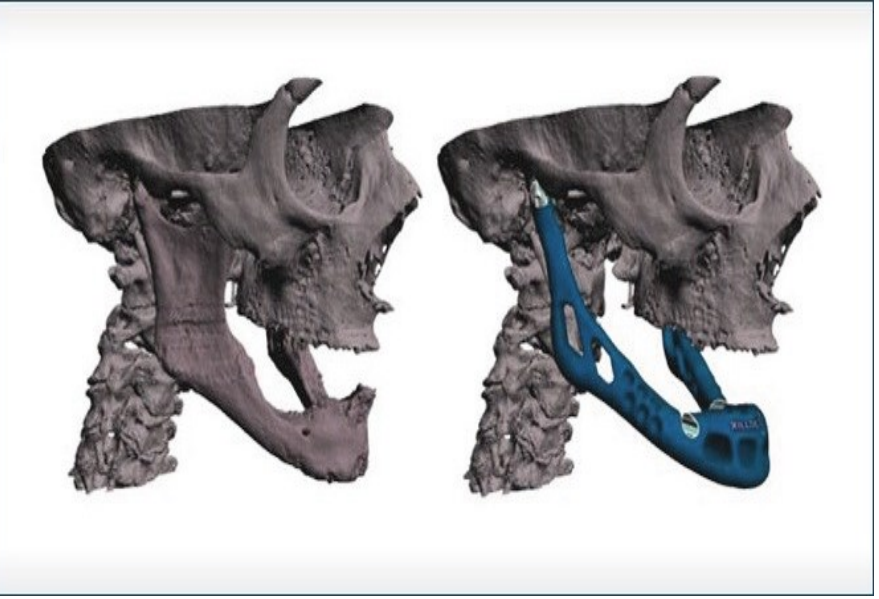
This originates from shellfish and can form hydrogels, which are three-dimensional networks that hold a lot of water. These are useful for wound healing, drug delivery, and cell encapsulation.



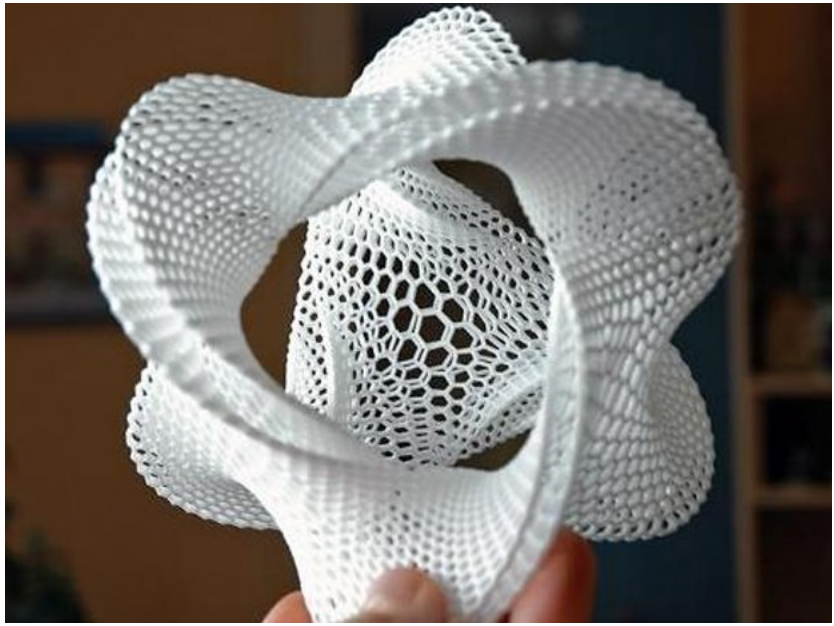
Hyaluronic Acid:

A naturally occurring sugar in our bodies, hyaluronic acid is used for dermal fillers, osteoarthritis treatment, and eye surgery due to its ability to lubricate and support tissues





Emerging Applications - 3D printing

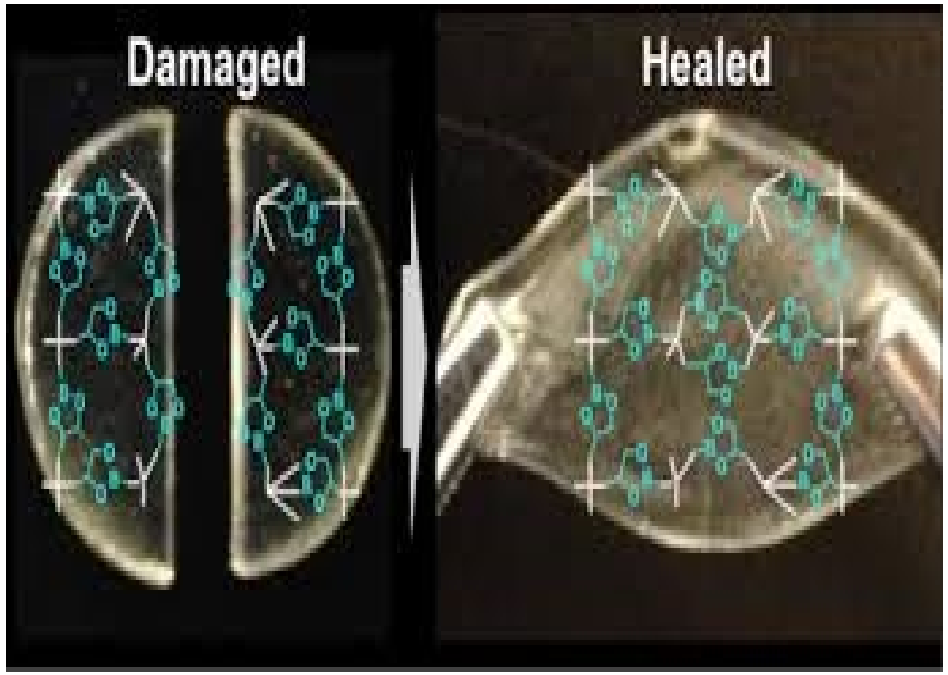


3D printing- Polyjet Printing

- PolyJet 3D printing is an additive manufacturing technology that uses a UV-curable liquid photopolymer, which produces high-resolution layers that enable the creation of complex and highly detailed 3D models
- PolyJet can produce full-color, flexible 3D printed materials, simulating the over-molding process in manufacturing.
- It creates realistic multi-material prototypes and models with full-color elements, labels, and true-to-life textures in a single operation.

3D Printed Polymers

- Polylactic Acid, or PLA, is beloved in the 3D printing community
- Acrylonitrile Butadiene Styrene (ABS) is a stalwart in the world of 3D printing.
- Thermoplastic Polyurethane, or TPU, is renowned for its elasticity and impact resistance, opening up a world of possibilities for 3D printing flexible, durable items
- Nylon, or Polyamide, is a staple in industries requiring high strength, durability, and resistance to wear and chemicals.
- Polycarbonate stands as a beacon of strength and clarity in the filament family, revered for its impact resistance and exceptional durability.



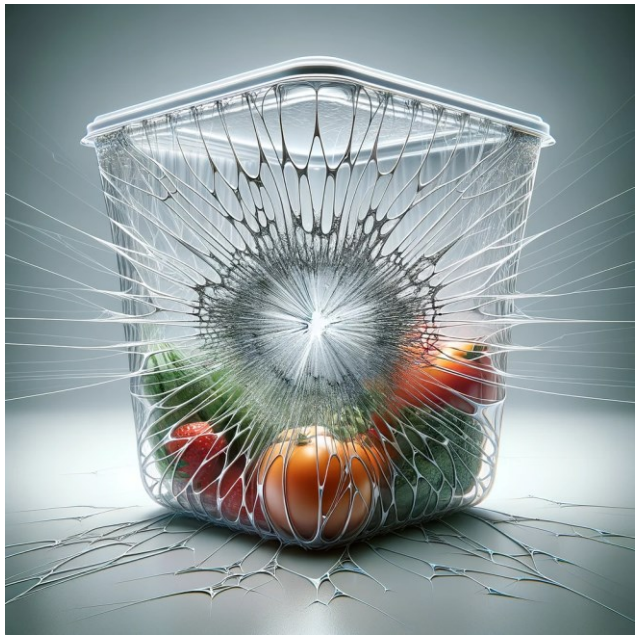
SELF-HEALING POLYMERIC MATERIALS

Self-healing polymeric materials

- Self-healing polymeric materials are advanced materials designed to automatically repair damage or degradation over time, without human intervention

<u>Applications</u>

- Automotive and Aerospace Industry
- Electronics and Packaging industry
- Medical Devices and Textiles industries



Significance of Healing efficiency

- **Healing Efficiency:** A measure of how well a self-healing material recovers its properties after damage.
- It is determined by comparing a key property of the healed material (e.g., tensile strength, electrical conductivity) to the same property of the undamaged material.

Self-healing material example

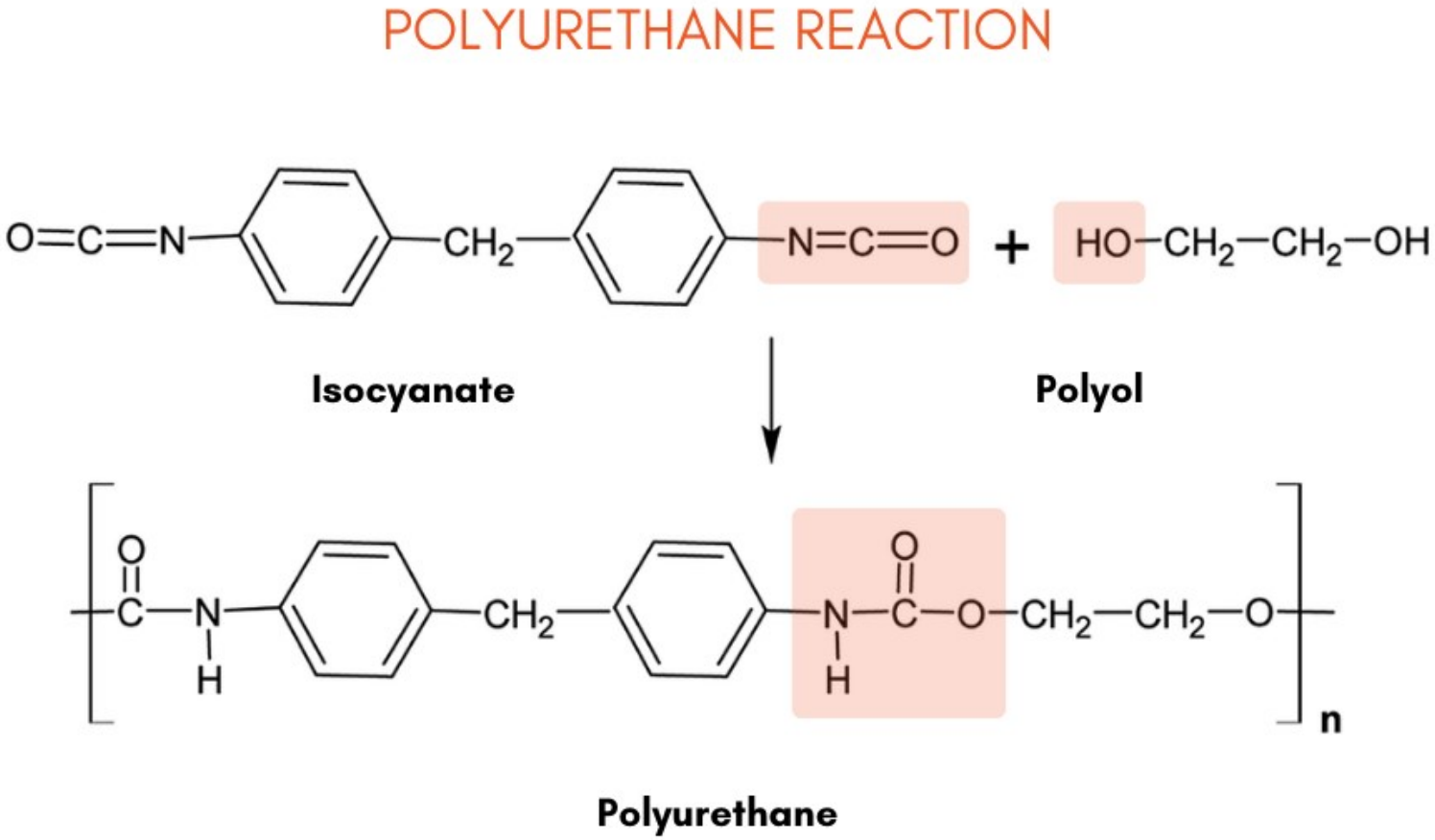
Polyurethane

Polyurethane is a type of polymer that is prepared by addition reaction between diisocyanate and diol or triol.

Application

- Polyurethanes (PU) are widely used in the automotive industry.
- Polyurethanes can be used as protective coatings to protect car surfaces from scratches, abrasion and corrosion.

Continued


POLYURETHANE REACTION
O=C=N
CH2
N=C=O
+
HO–CH2–CH2–OH
Isocyanate
Polyol
O
C
N
H
CH2
N
C
O
CH2
CH2
O
n
Polyurethane